Deep Learning HW1. Zhang Zeyu. 2026213422 the answers are highlighted by △

1.3.1

$\because \vec{s}=(2,1,-1,3)^T \quad \vec{q}=(1,\frac{1}{2},\frac{1}{2})^T \Rightarrow a_1=(2,1,-1)\begin{pmatrix}1\\ \frac{1}{2}\\ \frac{1}{2}\end{pmatrix}=2 \quad , \quad a_2=(1,-1,3)\begin{pmatrix}1\\ \frac{1}{2}\\ \frac{1}{2}\end{pmatrix}=2$

$\therefore \vec{h}=ReLU(\vec{a})=\begin{pmatrix}2\\ 2\end{pmatrix}$ (1)*

let $(\vec{v}_1,\vec{u}_1,\vec{f}_1),(\vec{v}_2,\vec{u}_2,\vec{f}_2)$ as the intermediate quantity of the experts, than:

$$\begin{cases}\vec{v}_1=W_{1,1}\vec{h}+\vec{b}_{1,1}=\begin{pmatrix}2\\ -\frac{3}{2}\end{pmatrix}, \ \vec{u}_1=\begin{pmatrix}2\\ 0\end{pmatrix}, \ \vec{f}_1=W_{2,1}\vec{u}_1+\vec{b}_{2,1}=\begin{pmatrix}2\\ \frac{5}{2}\end{pmatrix}\\ \vec{v}_2=W_{1,2}\vec{h}+\vec{b}_{1,2}=\begin{pmatrix}2\\ -1\end{pmatrix}, \ \vec{u}_2=\begin{pmatrix}2\\ 0\end{pmatrix}, \ \vec{f}_2=W_{2,2}\vec{u}_2+\vec{b}_{2,2}=\begin{pmatrix}\frac{1}{5}\\ \frac{2}{5}\\ 1\end{pmatrix}\end{cases} \quad (2)^*$$

the logits and gate weights of the router are calculated as:

$$\Rightarrow\begin{cases}\vec{q}=Softmax(G\vec{h})=Softmax\left(\begin{pmatrix}\frac{1}{4}&\frac{1}{4}\\ 0&0\end{pmatrix}\begin{pmatrix}2\\ 2\end{pmatrix}\right)=Softmax\left(\begin{pmatrix}0\\ 0\end{pmatrix}\right)=\begin{pmatrix}\frac{1}{2}\\ \frac{1}{2}\end{pmatrix}\\ \vec{r}=\vec{h}+\sum_{e=1}^{2}g_e\vec{f}_e=\vec{h}+g_1\vec{f}_1+g_2\vec{f}_2=\begin{pmatrix}\frac{69}{20}\\ \frac{69}{20}\end{pmatrix}, \ \because C\vec{r}+\vec{b}_c \Rightarrow\begin{pmatrix}\frac{1}{5}&0\\ 0&\frac{1}{5}\\ \frac{1}{5}&\frac{1}{5}\end{pmatrix}\begin{pmatrix}\frac{69}{20}\\ \frac{69}{20}\end{pmatrix}+\begin{pmatrix}\frac{69}{100}\\ \frac{69}{100}\\ \frac{98}{100}\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}\Rightarrow\hat{y}=\begin{pmatrix}\frac{1}{3}\\ \frac{1}{3}\\ \frac{1}{3}\end{pmatrix}\end{cases} \quad (3)^*$$

$$\mathcal{L}=-\frac{1}{m}\sum_{i=1}^{m}\sum_{k=1}^{k}y_k^{(i)}\log\hat{y}_k^{(i)} \xrightarrow{m=1,\,k=3} -\sum_{k=1}^{3}y_k\log\hat{y}_k \xrightarrow{\vec{y}=(0,1,0)^T} y_2\log\hat{y}_2=\log 3\approx 1.09861$$

1.3.2

let $z\in\mathbb{R}^K$ is the logits of sample $k\in\mathbb{R}^K$

let $p_i=\frac{e^{z_i}}{\sum_{j=1}^{K}e^{z_j}}$ is the probability after Softmax, than:

$$\because \begin{cases}\frac{\partial e^{z_i}}{\partial z_j}=\begin{cases}e^{z_i}, & i=j\\ 0, & i\neq j\end{cases}=\vec{1}[i=j]e^{z_i}\\ \frac{\partial z}{\partial z_j}=\frac{\partial}{\partial z_j}\sum_{c=1}^{K}e^{z_c}=e^{z_j}\end{cases}\Rightarrow\frac{\partial p_i}{\partial z_j}=\frac{\sum\frac{\partial e^{z_i}}{\partial z_j}-e^{z_i}\frac{\partial z}{\partial z_j}}{\sum^2}=\vec{1}[i=j]\frac{e^{z_i}}{\sum}-\frac{e^{z_i}}{\sum}\cdot\frac{e^{z_j}}{\sum}=p_i(\vec{1}[i=j]-\vec{p}_j)=\begin{cases}p_i(1-p_i), & i=j\\ -p_ip_j, & i\neq j\end{cases}$$

let Jacobian matrix $\vec{J}_{ij}=\frac{\partial p_i}{\partial z_j}\in\mathbb{R}^{K\times K}(i:1\to K,j:1\to K)\Rightarrow \vec{J}=diag(\vec{p})-\vec{p}\vec{p}^T$

$$\therefore \begin{cases}\vec{J}^T=\vec{J} \ ①\\ \vec{p}^T\vec{1}=\sum_i\vec{p}_i=1\Rightarrow\vec{J}\vec{1}=diag(\vec{p})\vec{1}-\vec{p}(\vec{p}^T\vec{1})=\vec{p}-\vec{p}=\vec{0} \ ②\end{cases}$$

let cross-entropy $\ell=-\sum_{a=1}^{K}y_a\log p_a$, $\frac{\partial\ell}{\partial z_b}=\sum_{a=1}^{K}\frac{\partial\ell}{\partial p_a}\frac{\partial p_a}{\partial z_b}=\sum_a(-\frac{y_a}{p_a})p_a(\vec{1}[a=b]-p_b)=-\sum_a y_a\vec{1}[a=b]+\sum_a y_ap_b$

$$\frac{\sum_a y_a\vec{1}[a=b]=y_b}{\sum_a y_ap_b=p_b\sum_a y_a}\Rightarrow\frac{\partial\ell}{\partial z_b}=-y_b+p_b\sum_a y_a\xrightarrow{\sum_a y_a=1}\frac{\partial\ell}{\partial z_b}=p_b-y_b, \ \nabla_z\ell=p-y\in\mathbb{R}^K \ ③$$

when the mini batch enters. let the batch has m samples:

the average loss $\mathcal{L}=\frac{1}{m}\sum_{i=1}^{m}\ell^{(i)}\Rightarrow\nabla_{z^{(i)}}\mathcal{L}=\frac{1}{m}\nabla_{z^{(i)}}\ell^{(i)}\overset{③}{=\!=}(p^{(i)}-y^{(i)})\frac{1}{m}$, so there's no need for another average.

1.3.3

for general m, D, H, E, K, let $\vec{t}^{(i)} = G\vec{h}^{(i)}$, $\vec{z}^{(i)} = C\vec{r}^{(i)} + b_c$ for sample i,

let $\beta^{(i)} := C^T\vec{\alpha}^{(i)} \in \mathbb{R}^D$. ① then $\nabla_C \mathcal{L} = \sum_i \vec{\alpha}^{(i)}(\vec{r}^{(i)})^T \in \mathbb{R}^{K\times D}$, $\nabla_{b_c}\mathcal{L} = \sum_i \vec{\alpha}^{(i)} \in \mathbb{R}^K$

for expert e, $r_d = h_d + \sum_e g_e f_{e,d} \Rightarrow \frac{\partial r_d}{\partial f_{e,j}} = g_e \mathbb{I}[d=j] \Rightarrow \frac{\partial \mathcal{L}}{\partial f_{e,j}} = \sum_{d=1}^{D} \frac{\partial \mathcal{L}}{\partial r_d}\frac{\partial r_d}{\partial f_{e,j}} = \sum_d \beta_d g_e \mathbb{I}[d=j] = g_e\beta_j$

$\therefore \eta_e = \nabla_{f_e}\mathcal{L} = g_e\vec{\beta} \in \mathbb{R}^D$

$\vec{f} = W_2\vec{u} + b_2 \Rightarrow f_d = \sum_{j=1}^{H}(W_2)_{dj}u_j + (b_2)_d$ (the 2nd layer) $\xrightarrow{\frac{\partial f_d}{\partial (W_2)_{dj}} = u_j}$ $\frac{\partial \mathcal{L}}{\partial (W_2)_{dj}} = \frac{\partial \mathcal{L}}{\partial f_d}\cdot\frac{\partial f_d}{\partial (W_2)_{dj}} = \eta_d u_j = (\vec{\eta}\vec{u}^T)_{dj}$

$\Rightarrow \frac{\partial \mathcal{L}}{\partial u_j} = \sum_{d=1}^{D}\frac{\partial \mathcal{L}}{\partial f_d}\frac{\partial f_d}{\partial u_j} = \sum_d \eta_d (W_2)_{dj} \Rightarrow \nabla_u \mathcal{L} = W_2^T\vec{\eta}$

$\because ReLU'(0) = 0 \ \therefore \frac{\partial u_j}{\partial v_j} = \mathbb{I}[v_j > 0] \Rightarrow \frac{\partial \mathcal{L}}{\partial v_j} = \frac{\partial \mathcal{L}}{\partial u_j}\mathbb{I}[v_j>0] \Rightarrow \delta := \nabla_v \mathcal{L} = (W_2^T\vec{\eta}) \odot \vec{\mathbb{I}}[v>0] \in \mathbb{R}^H$

$\vec{v} = W_1\vec{h} + b_1 \Rightarrow v_j = \sum_{d=1}^{D}(W_1)_{jd}h_d + (b_1)_j \Rightarrow \frac{\partial \mathcal{L}}{\partial (W_1)_{jd}} = \delta_j h_d$, $\frac{\partial \mathcal{L}}{\partial (b_1)_j} = \delta_j$ (the 1st layer)

$\therefore \nabla_{W_{2,e}}\mathcal{L} = \sum_i \eta_e^{(i)}(u_e^{(i)})^T \in \mathbb{R}^{D\times H}$, $\nabla_{b_{2,e}}\mathcal{L} = \sum_i (\eta_e^{(i)}) \in \mathbb{R}^D$, $\nabla_{W_{1,e}}\mathcal{L} = \sum_i \delta_e^{(i)}(h^{(i)})^T \in \mathbb{R}^{H\times D}$, $\nabla_{b_{1,e}}\mathcal{L} = \sum_i (\delta_e^{(i)}) \in \mathbb{R}^H$

$\Rightarrow \nabla_{h^{(i)}}\mathcal{L}|_{EP} = \sum_e W_{1,e}^T\delta_e^{(i)}$ ②

for router path, $t = Gh$, $\frac{\partial \mathcal{L}}{\partial h_d}|_{RP} = \sum_{e=1}^{E}\frac{\partial \mathcal{L}}{\partial t_e}\frac{\partial t_e}{\partial h_d} = \sum_e \rho_e G_{ed}$, $\rho_e := g_e^{(i)}\left(\gamma_e^{(i)} - \sum_j q_j^{(i)}\gamma_j^{(i)}\right)$, $\gamma_e^{(i)} := (f_e^{(i)})^T\beta^{(i)} \in \mathbb{R}$ ③

$\because r(h) = \vec{h} + \sum_e g_e(h) f_e(h)$, $\therefore dr = dh + \sum_e g_e\, df_e + \sum_e f_e\, dg_e \Rightarrow \nabla_{h^{(i)}}\mathcal{L} = \beta^{(i)} + \sum_e W_{1,e}^T\delta_e^{(i)} + G^T\rho^{(i)}$

(residual, expert path, route path)

$\because \vec{\alpha} = \begin{pmatrix}1/3\\-2/3\\1/3\end{pmatrix}$, $\vec{\beta} = \begin{pmatrix}2/\sqrt{5}\\-1/\sqrt{5}\end{pmatrix}$, $\therefore \vec{\gamma} = \begin{pmatrix}1/10\\1/2\end{pmatrix}$, $\vec{\rho} = \begin{pmatrix}1/2\\1/2\end{pmatrix} \odot [\vec{\gamma} - 3/100\,(1)] = \begin{pmatrix}-1/200\\1/200\end{pmatrix}$ $\therefore \frac{\partial \mathcal{L}}{\partial G_{1,1}} = \rho_1 h_1 = -\frac{1}{100}$

1.3.4

let $\delta_j^{(i)} = \frac{\partial \mathcal{L}}{\partial a_j} = (\nabla_{h^{(i)}}\mathcal{L})_j \mathbb{I}[a_j^{(i)} > 0]$ $\because a_j^{(i)} = \sum_{t=1}^{3} q_t s_{j+t-1}^{(i)}$

the gradient of shared kernel $\frac{\partial \mathcal{L}}{\partial q_t} = \sum_{i=1}^{m}\sum_{j=1}^{2}\delta_j^{(i)} s_{j+t-1}^{(i)}$, $t = 1,2,3$.

for single enter location l, $\frac{\partial \mathcal{L}}{\partial s_l^{(i)}} = \sum_{1\le j\le 2,\ 1\le l-j+1\le 3}\delta_j^{(i)} q_{l-j+1}$, $\because \delta = \nabla_h \mathcal{L} = (369, -173)^T/2400$

$\nabla_q \mathcal{L} = \frac{1}{2400}\left[369\begin{pmatrix}2\\1\\-1\end{pmatrix} - 173\begin{pmatrix}1\\-1\\2\end{pmatrix}\right] = \frac{1}{2400}\begin{pmatrix}525\\521\\-868\end{pmatrix} = \begin{pmatrix}7/32\\87/400\\-217/600\end{pmatrix}$, $\frac{\partial \mathcal{L}}{\partial s_2} = \frac{369}{2400}\cdot\frac{1}{2} - \frac{173}{2400}\cdot 1 = \frac{1}{1600}$

give learning rate $\eta = 0.1$, $q_{new} = \begin{pmatrix}1/2\\1/2\\1/2\end{pmatrix} - \frac{1}{10}\begin{pmatrix}7/32\\87/400\\-217/600\end{pmatrix} = \begin{pmatrix}313/320\\1913/2400\\3217/3600\end{pmatrix} \approx (0.978125, 0.797500, 0.893611)^T$.